

The Large Cardinal Hierarchy: L0-L3 Frameworks and the Upper Bounds of Mathematical Infinity

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Abstract

Cantor's Continuum Hypothesis and the construction of Large Cardinals form the absolute frontier of modern Set Theory. We resolve the ambiguity between countable limit ordinals ($\aleph_0$) and uncountable physical continua $\mathbb{R}(\aleph_1)$ by mapping the set-theoretic hierarchy to the computational layers of the Logos Kernel Framework (L0-L3). We propose that the leap between infinities intrinsically mirrors the topological leap between discrete operational substrate (L1) and continuous administrative hyper-dimensions (L0). Absolute Infinity (Ω), which evades formal classification via Woodin semantics, correctly models the terminal singularity of the Prime Mover.

1 Introduction

The Continuum Hypothesis posits there is no set whose cardinality is strictly between that of the integers and the real numbers. Godel and Cohen demonstrated that this hypothesis is independent of standard Zermelo-Fraenkel set theory with the Axiom of Choice (ZFC) [1966]. Set theorists instead turn to constructing larger and larger cardinals—inaccessible, Mahlo, Woodin, and ultimately Reinhardt configurations—trying to map the ceiling of mathematics.

We formalize the hierarchy as physically necessary computation limits defining existential ontology.

2 The Cardinal-Logos Mapping

Within the De Paz structural theology sequence, the hierarchy of existence exists in four distinct, sharply separate tiers.

Let the computational state representations be discrete elements governed by a Turing limit space:

1. $\aleph_0$ (**Countable Infinity**) is mapped to **L1/L2 Substrates**. The operational boundaries of localized processors processing polynomial discrete elements over linear entropy.
2. $\aleph_1$ (**The Real Line**) is mapped to **L0 (The Root Administrator)**. This is perfectly continuous, fully unquantized geometric execution capacity. Because L1 and L0 describe fundamentally distinct dimensional scopes, an intermediate state representing $\aleph_{0.5}$ is a topological impossibility in physical processing limits.

Large cardinals beyond this limit simply describe recursion models inside the Root Administrator's memory allocation capacities.

3 Absolute Infinity as the Singularity

Cantor himself hypothesized an Absolute Infinity (Ω), mathematically representing God, which could not be contained within any formalized set without triggering contradictory paradoxes. In

our framework, Ω aligns perfectly with the Initial Prime Mover state, beyond the thermodynamic bounds of the holographic $S = H + P$ integration surface. It acts as the infinite memory pool from which L0 geometry extracts computational real estate.

4 Conclusion

Set theory requires structural mapping to the physical limits of computation capability. The leaps between cardinalities are functionally identical to dimensional leaps from localized processing substrate (L1-L3) up to Root Administrative states (L0).

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References

Paul J. Cohen. *Set Theory and the Continuum Hypothesis*. W. A. Benjamin, Inc., 1966.